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Instructions

1. The exam duration is 40 minutes, starting at 12:10. Ensure that you submit both the question paper and the answer book by 12:50. Please be aware that **late submissions** may incur **penalties**.

2. Read all questions carefully. Misinterpretations will **not** be considered. It is essential to write your solutions clearly and legibly in the answer book. In a previous quiz, a few students submitted answers that were not clearly written. Going forward, such responses will **not** be evaluated.

3. Any form of unfair practice will result in an **F grade** and will be reported to the disciplinary committee.

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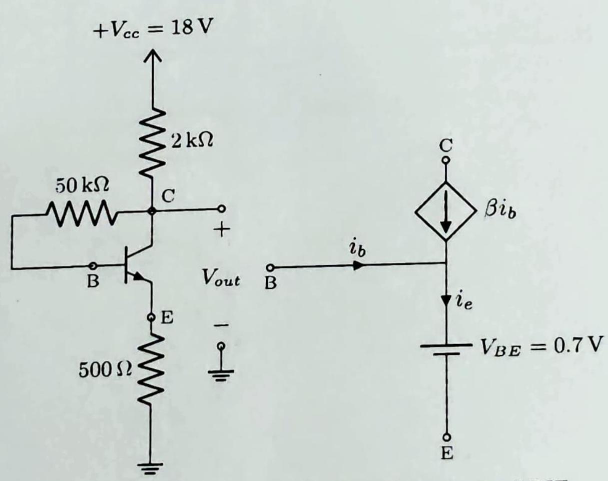
1. (3 points) Consider the BJT circuit shown in Fig. 1(a). Assume that the BJT is operating as an amplifier, and the equivalent circuit during this mode is shown in Fig. 1(b). Further, consider the following parameters: $\beta = 200$ and $V_{BE} = 0.7\,\text{V}$. Compute

   (a) (2 points) The current drawn from the supply $+V_{CC}$ (in mA) ________.

   (b) (1 point) The voltage V_{CE} ________.



(a) Circuit (b) Equivalent circuit of BJT

Fig. 1: Circuit with BJT (Questions 1)

2. (4 points) Using the virtual short concept, compute the output voltage v_o for the Op-Amp circuit shown in Fig. 2. Write your answer here: - 7.5 V.

3. (3 points) For the circuit shown in Fig. 3, compute the following quantities at port a-o:

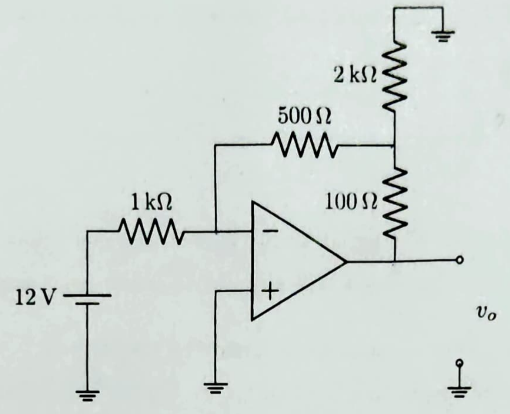


Fig. 2: Circuit for question 2

(a) (1 point) The open-circuit voltage at port a-o. 28.571 V

(b) (1 point) The short-circuit current at port a-o. 28.6 A

(c) (1 point) The equivalent (output) resistance at port a-o. 0.998 Ω

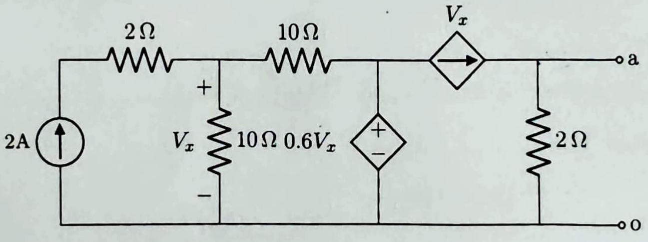


Fig. 3: Circuit for question 3